

Daniel Villavicencio

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Software engineer with hands-on experience in systems-level development, CI/CD automation, and infrastructure tooling. Proficient in Python, Java, C/C++, and Bash across Linux environments. Familiar with AI/ML frameworks and authentication protocols through coursework and independent study.

Technical Skills

Programming Languages: Java, Python, C/C++, Go, Lua, Bash, SQL

Databases: PostgreSQL, MySQL, SQLite, Redis, Firestore

Infra & DevOps: Git, Docker, Kubernetes, Linux, AWS, GCP, TensorFlow, GitHub Actions, Jira, Prometheus, RabbitMQ, Loki, Gitlab Pipelines

Concepts: Object-Oriented Design, Distributed Systems, Scripting & Automation, Software Debugging, Secure Coding Practices, Security & Auth Protocols, AI/ML, IaC

Experience

United Natural Foods (UNFI) | *Software Engineer I* | Remote | June 2022 - October 2023

- Designed and executed unit and integration test suites on Linux-based systems using established SDLC methodologies, partnering with cross-functional teams to debug and resolve production defects.
- Developed and maintained high-performance REST APIs across multiple Java/Spring Boot microservices in a Linux environment, applying secure coding practices and authentication best practices.
- Contributed to a legacy modernization initiative, migrating 25% of services to a modern Kubernetes architecture while maintaining CI/CD pipeline integrity via GitLab.

Open Source Contributions

Moby (Docker Engine) | Jan 2026 - Present

- **Test Infrastructure Modernization:** Modernized legacy integration tests in Docker Engine's Go test suite, directly improving developer tooling reliability and maintainability.
- **Systems Debugging:** Resolved flaky test failures through deep analysis of Linux kernel-level networking behaviors, applying advanced debugging techniques to identify and eliminate resource inefficiencies.
- **Cross-Platform Validation:** Designed test cases ensuring behavioral parity across diverse OS environments, navigating user namespaces and unprivileged container execution.

Docker Docs | May 2025 - Present

- **Architectural Research:** Independently discovered and validated undocumented host network interface access patterns in Docker Swarm, reproducing the behavior through targeted YAML configuration testing and opening a documentation PR to capture findings for the community.

Projects

Go Schema Migration Tool (GSMT) | Jun 2025 - Present

- Developed a lightweight Go library for automated SQL schema migration management, designed for in-code integration without external tooling dependencies, with multi-dialect support for PostgreSQL and SQLite3.

Rivenbot Suite | Oct 2023 – May 2025

- Architected and evolved a Destiny 2 player statistics platform from a monolithic Java/Spring WebFlux Discord bot into a distributed suite of Go microservices, including an API gateway with rate-limiting, a RabbitMQ-driven crawler pipeline, and a PGCR processing service backed by PostgreSQL and Redis.
- Engineered a Docker Swarm deployment stack with reusable GitHub Actions workflows, applying infrastructure-as-code practices for automated service orchestration across Digital Ocean.
- Implemented end-to-end observability across the full suite using Loki and Grafana to monitor application performance and service health.
- Validated service reliability across both generations of the platform — JUnit, Mockito, Testcontainers, and WireMock for the Java monolith, and Go's native testing framework with Testcontainers for the microservices rewrite.

PGCR Batch Processor | Nov 2024 - May 2025

- Built a Java/Spring Batch pipeline to crawl and process 2–3TB of compressed Post Game Carnage Report data from Bungie's Destiny 2 API, condensing raw payloads into a structured PostgreSQL database for downstream analytics.
- Designed a chunk-oriented batch processing architecture with discrete steps to efficiently handle high-volume concurrent data ingestion backed by Redis.
- Instrumented the pipeline with Prometheus and Grafana to monitor batch job execution and data processing performance.

Certifications

AWS Cloud Practitioner | Amazon Web Services | May 2025 - May 2028

Education

California State University, Sacramento

Bachelor of Science, Computer Science | GPA: 3.77/4.00 | Dean's Honor Roll

Relevant Coursework: Operating Systems, Algorithms & Data Structures, Intelligent Systems.